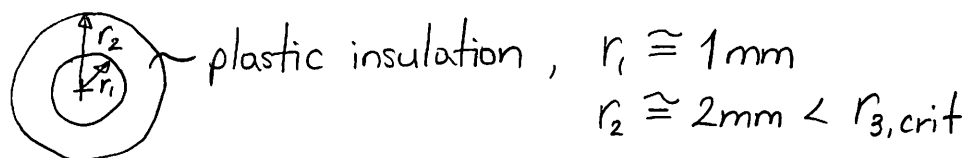


In general: $k_i \approx 0.1 \text{ W/m}\cdot\text{K}$ (common insulating material)
 $h_o \approx 5 \text{ W/m}^2\cdot\text{K}$ (natural convection)
 $r_{3,\text{crit}} \approx 1 \text{ cm} \Rightarrow$ We typically design our systems (HVAC&R) to be larger than this, hence OK to insulate.

∴ We can insulate hot water and steam pipes without worrying about increasing external heat transfer losses

So how about electrical wires?

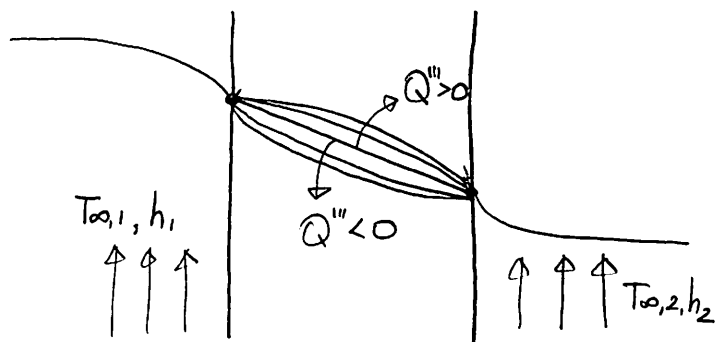


Although $r_2 < r_{3,\text{crit}}$, we typically want to cool our electrical wires & NOT thermally insulate them, so using a bigger or thicker insulation with a lower R_{TOT} is good!

Heat Generation

Assumptions:

- 1) 1D
- 2) Steady State
- 3) Constant properties
- 4) Uniform Q''' ($Q''' = \text{constant}$)



Note, the heat transfer rate and heat flux now need not be constant with respect to x .

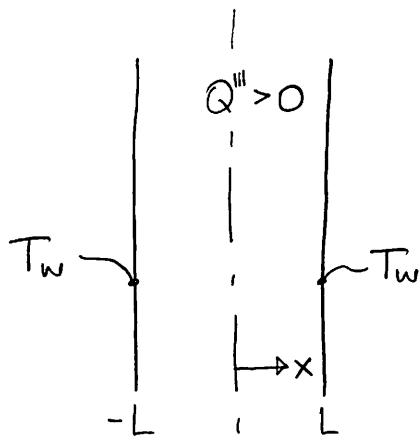
Writing out our heat equation: (in cartesian)

$$\frac{\partial^2 T}{\partial x^2} + \underbrace{\frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}}_{O(1D)} + \frac{Q'''}{k} = \underbrace{\frac{\rho c_p}{k} \frac{\partial T}{\partial t}}_{O(SS)}$$

So our equation becomes:

$$\frac{\partial^2 T}{\partial x^2} + \frac{Q'''}{k} = 0 \quad (1)$$

Let's do the simplest case BC's (1st kind)



$$T(x=L) = T_w$$

$$T(x=-L) = T_w$$

$$\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0 \quad (\text{Symmetry or insulated})$$

↳ According to symmetry, the temperatures to the left and right of the plane of symmetry are the same, hence $\partial T / \partial x = 0$.

Integrating (1):

$$\frac{\partial^2 T}{\partial x^2} = -\frac{Q'''}{k}$$

$$\int \frac{\partial^2 T}{\partial x^2} dx = \int -\frac{Q'''}{k} dx$$

$$\frac{\partial T}{\partial x} = -\frac{Q'''}{k} x + C_1$$

But we know $\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0 \Rightarrow C_1 = 0$

Integrating once more: $T(x) = -\frac{Q''' x^2}{2k} + C_2$

We know $T(x=\pm L) = T_w$

$$T(x=L) = T_w = -\frac{Q'''L^2}{2k} + C_2 \Rightarrow C_2 = T_w + \frac{Q'''L^2}{2k}$$

$$\boxed{T(x) = T_w + \frac{Q'''L^2}{2k} \left(1 - \left(\frac{x}{L}\right)^2\right)} \Rightarrow \text{Temperature profile inside the slab.}$$

So where is $T_{\max}$?

$$\frac{\partial T}{\partial x} = -\frac{Q'''x}{k} = 0 \Rightarrow x=0 \text{ is the solution}$$

Let's check if it is a maximum at $x=0$:

$$\left. \frac{\partial^2 T}{\partial x^2} \right|_{x=0} = -\frac{Q'''}{k} < 0 \text{ (So } T \text{ is a max at } x=0)$$

$$\boxed{T_{\max} = T_w + \frac{Q'''L^2}{2k}}$$

How about heat flux q'' :

$$q'' = -k \frac{\partial T}{\partial x} = Q'''x = \boxed{q'' = Q'''x}$$

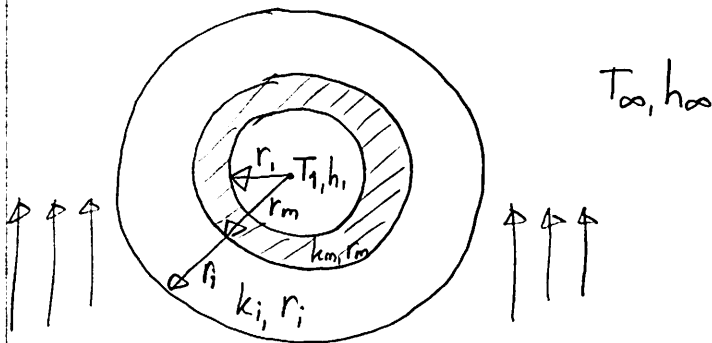
At the boundaries: ($x=\pm L$) $\boxed{q''|_{x=L} = Q'''L} \Rightarrow \text{makes sense.}$

Other Cases:

If we had different wall temperatures: $T(x=-L) = T_{s1}$
 We would solve the same problem but $T(x=L) = T_{s2}$
 with the new B.C.'s.

$$\boxed{T(x) = \frac{Q'''L^2}{2k} \left(1 - \left(\frac{x}{L}\right)^2\right) + \frac{T_{s2} - T_{s1}}{2} \cdot \frac{x}{L} + \frac{T_{s1} + T_{s2}}{2}}$$

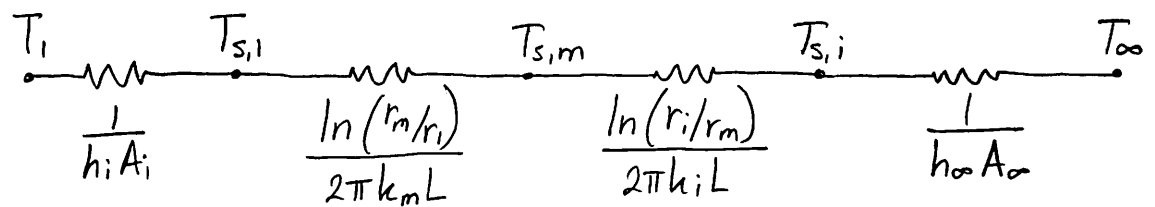
Example / Pipe flow & freezing water



$$\begin{aligned} T_i &= 10^\circ\text{C}, h_i = 100\text{W/m}^2\cdot\text{K} \\ r_i &= 2\text{cm} \\ r_m &= 2.5\text{cm}, k_m = 400\text{W/m}\cdot\text{K} \\ r_i &= 5\text{cm}, k_i = 2\text{W/m}\cdot\text{K} \\ T_\infty &= ?, h_\infty = 50\text{W/m}^2\cdot\text{K} \end{aligned}$$

At what outside temperature T_∞ , will the water in the pipe begin to freeze?

We can solve this with our thermal resistance approach



We know when freezing starts that $T_{s,i} = 0^\circ\text{C} \Rightarrow$ freezing point of water.

Let's solve the first leg of the resistance diagram:

$$Q = \frac{\Delta T}{R} = \frac{T_i - T_{s,i}}{\frac{1}{h_i A_i}} = \frac{10^\circ\text{C} - 0^\circ\text{C}}{\frac{1}{(100\text{W/m}^2\cdot\text{K})(2\pi(0.02\text{m})(1\text{m}))}}$$

$$Q = 125.6\text{W}$$

A_i (assuming 1m long tube, doesn't matter in the end)

Since we don't have any heat generation, we know that Q is constant in our thermal circuit. So now we can solve for T_∞ .

$$\begin{aligned}
 R_{TOT} &= \frac{1}{h_i A_i} + \frac{\ln(r_m/r_i)}{2\pi k_m L} + \frac{\ln(r_i/r_m)}{2\pi h_i L} + \frac{1}{h_\infty A_\infty} \\
 &= \frac{1}{(100)(2\pi(0.02)(1))} + \frac{\ln(2.5/2)}{2\pi(400)(1)} + \frac{\ln(5/2.5)}{2\pi(1.2)(1)} + \frac{1}{(50)(2\pi(0.05)(1))} \\
 &= 0.0796 \text{ K/W} + \underbrace{8.88 \times 10^{-5} \text{ K/W}}_{\text{Thermal resistance of metal}} + \underbrace{0.055 \text{ K/W}}_{\text{Thermal resistance of insulation}} + 0.064 \text{ K/W}
 \end{aligned}$$

$$R_{TOT} = 0.20 \text{ K/W}$$

Now we can solve for T_∞ :

$$Q = \frac{\Delta T_{TOT}}{R_{TOT}} = 125.6 \text{ W} = \frac{10^\circ\text{C} - T_\infty}{0.2 \text{ K/W}}$$

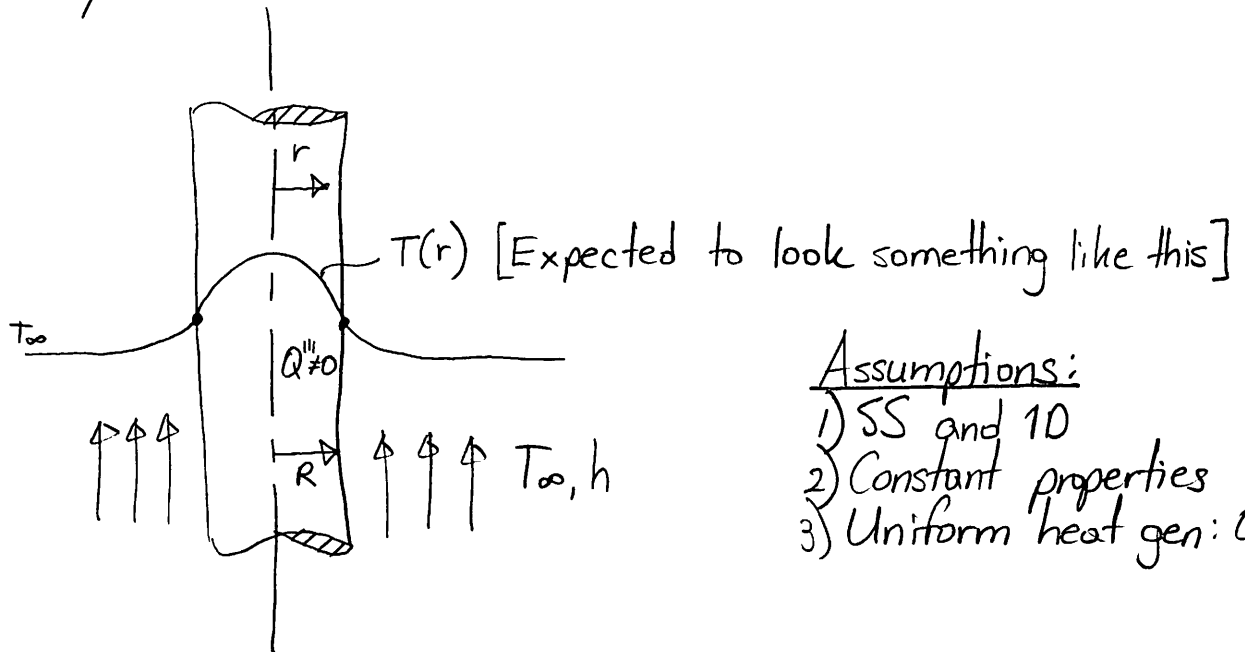
$$10^\circ\text{C} - T_\infty = 24.98^\circ\text{C}$$

$$\boxed{T_\infty = -15^\circ\text{C}}$$

Therefore, we don't need to worry about our pipe freezing until the outside air reaches -15°C . In reality, we probably should add some more insulation just in case since on occasion, temperatures in Illinois can reach below -15°C .

Circular Cylinder with Heat Generation

Many times the constant boundary temperature condition is not valid. Here, we will explore this case in a radial geometry:



Assumptions:

- 1) SS and 1D
- 2) Constant properties
- 3) Uniform heat gen: $Q''' = \text{const}$

Writing out our heat equation in radial form:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(k r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(k r \frac{\partial T}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + Q''' = \rho c_p \frac{\partial T}{\partial t}$$

1D
1D
SS

$$\frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + Q''' = 0$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = - \frac{Q'''}{k}$$

$$\frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = - \frac{Q''' r}{k} \quad (\text{Integrate wrt. } r)$$

$$\int \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = - \frac{Q'''}{k} \int r \, dr$$

$$r \frac{\partial T}{\partial r} = - \frac{Q''' r^2}{2k} + C_1 \quad (\text{Integrate again})$$

$$\int 2T = -\frac{Q'''}{2k} \int r dr + \int \frac{C_1}{r} dr$$

$$T(r) = -\frac{Q''' r^2}{4k} + C_1 \ln r + C_2$$

Our boundary conditions are a little different now:

$$1) -k \frac{\partial T}{\partial r} \Big|_{r=R} = h (T \Big|_{r=R} - T_\infty) \quad [\text{Energy balance at the surface}]$$

$$2) \frac{\partial T}{\partial r} \Big|_{r=0} = 0 \quad [\text{Symmetry of the problem}]$$

Right away we can solve for one of our constants. Since $\ln r = \infty$, then $C_1 = 0$ since we know we have a finite temperature.

The other way to see this is to apply B.C. #2 to our solution:

$$\frac{\partial T}{\partial r} \Big|_{r=0} = -\frac{Q'''(0)}{2k} + \frac{C_1}{0} \rightarrow \infty, \text{ hence the only way this works is if } C_1 = 0$$

Now we apply BC #1 to our equation:

$$T(r) = -\frac{Q''' r^2}{4k} + C_2$$

$$-k \frac{\partial T}{\partial r} \Big|_{r=R} = +\frac{k Q''' R}{2k} = h \left(-\frac{Q''' R^2}{4k} + C_2 - T_\infty \right) \quad [\text{BC \#1}]$$

$$\frac{1}{h} \left[\frac{k Q''' R}{2k} + \frac{h Q''' R^2}{4k} + h T_\infty \right] = C_2$$

$$\frac{1}{h} \left[\frac{Q''' R}{2} \left(1 + \frac{hR}{2k} \right) + h T_\infty \right] = C_2 \Rightarrow \text{Done.}$$